

Realization Spaces of NEO Phased Matroids



國立陽明交通大學
NATIONAL YANG MING CHIAO TUNG UNIVERSITY

Yu-Chuan Yu & Advisor: Prof. Chi Ho Yuen

Institute of Applied Mathematics, National Yang Ming Chiao Tung University

Introduction

Realization spaces encode the rich topology of combinatorial geometries.

- **Realization space:** the set of all matrices (or subspaces) realizing a fixed oriented matroid (OM).
- **Mnëv Universality thm. (1985):** Realization spaces of OMs can be stably equivalent to any semialgebraic set [1].
- **Phased Matroids:** Complex analogue of OMs.
- **Ruiz (2013):** Uniform not essentially oriented (NEO) phased matroids have singleton realization spaces [2].

Topology of Realization Spaces

Class	$\tilde{\mathcal{R}}_{\mathbb{C}}(\mathcal{M})$	Reference
OM / EO	universality	Mnëv (1985)
Uniform NEO	singleton	Ruiz (2013)
NEO with rank+2 uniform restriction	singleton	our result

Central Question

What minimal combinatorial conditions guarantee singleton (or contractible) realization spaces for NEO phased matroids?

Phased Matroids and Realizations

- A **phirotope** of rank d is a nonzero alternating map

$$\varphi : \mathbb{E}^d \rightarrow \mathbb{S}^1 \cup \{0\}, \quad \mathbb{S}^1 = \{z \in \mathbb{C} \mid |z| = 1\},$$

satisfying the **phase Grassmann-Plücker relation**: For all $e_{\bullet} \in \mathbb{E}^{d+1}$, $f_{\bullet} \in \mathbb{E}^{d-1}$

$$0 \in \text{ph} \left(\text{span}_{\mathbb{R}_{>0}} \left\{ (-1)^k \varphi(\widehat{e}_k) \cdot \varphi(e_k, f_{\bullet}) \right\}_{k=1}^{d+1} \right).$$

- A **phased matroid** is a pair $\mathcal{M} = (\mathbb{E}, [\varphi])$, where $[\varphi]$ is the equivalence class of phirotopes under global scaling:

$$\varphi \sim \alpha \cdot \varphi \quad \text{for all } \alpha \in \mathbb{S}^1.$$

- A **realization** of $\mathcal{M}$ is a full-rank matrix $M = (\vec{v}_e)_{e \in \mathbb{E}} \in \mathbb{C}^{d \times \mathbb{E}}$, such that the map

$$(e_1, \dots, e_d) \mapsto \text{ph}(\det(\vec{v}_{e_1}, \dots, \vec{v}_{e_d})) \quad \text{is in } [\varphi].$$

- Realization space is closed under:

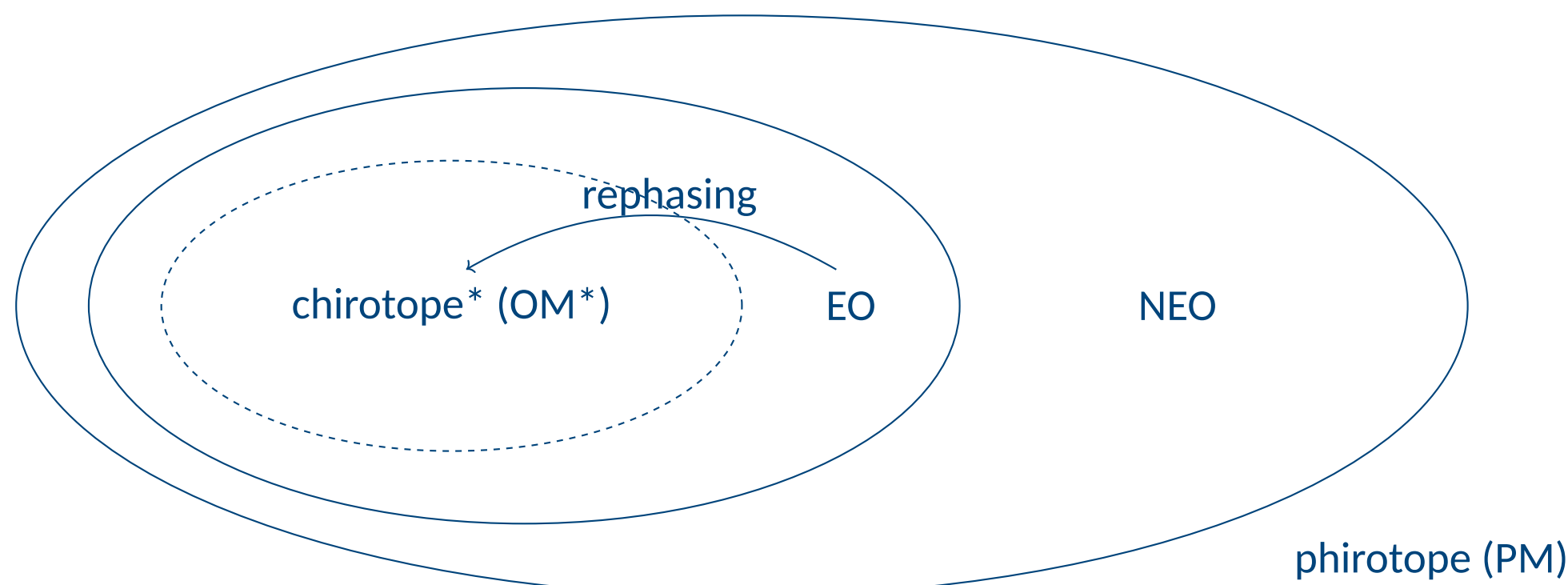
- $\text{GL}(d, \mathbb{C})$: change of basis (left multiplication),
- $(\mathbb{R}_{>0})^{\mathbb{E}}$: positive column rescaling.

Rephasing and EO/NEO Dichotomy

A **rephasing** is a function $\beta : \mathbb{E} \rightarrow \mathbb{S}^1$, acting on a phirotope φ by

$$\varphi^{\beta}(e_{\bullet}) := \left(\prod_i \beta(e_i) \right) \cdot \varphi(e_{\bullet}).$$

- **Essentially Oriented (EO):** There exists a rephasing β such that $\text{Im}(\varphi^{\beta}) \subseteq \{0, \pm\alpha\}$, for some $\alpha \in \mathbb{S}^1$.
- **Not Essentially Oriented (NEO):** No such β exists.

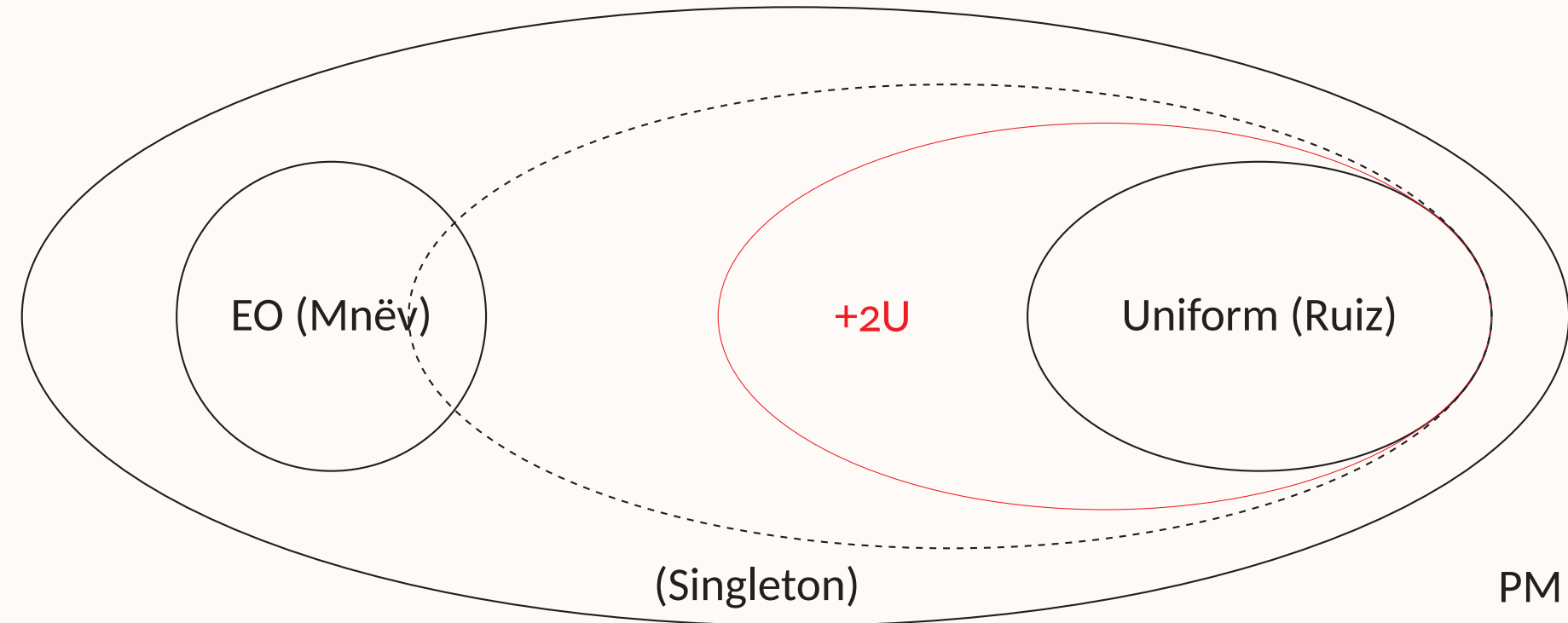


Reference

- [1] Jürgen Richter-Gebert. Mnëv's universality theorem revisited. *Sém. Lothar. Combin.*, 34:B34h, 1995.
- [2] Amanda Ruiz. *Realization spaces of phased matroids*. PhD thesis, 2013. State University of New York at Binghamton.

Main Result (NEO+2U Criterion)

Let $\mathcal{M}$ be a realizable NEO phased matroid of rank d . If some $d+2$ elements form a uniform restriction, then $\tilde{\mathcal{R}}_{\mathbb{C}}(\mathcal{M})$ is a singleton.



Framework for Analyzing Realization Spaces

$$\begin{array}{ccccc} \{M \text{ realizes } \mathcal{M}\} & \xrightarrow{\text{GL} \setminus -} & \mathcal{R}(\mathcal{M}) & \xrightarrow{-/\mathbb{R}_{>0}^{\mathbb{E}}} & \tilde{\mathcal{R}}(\mathcal{M}) \\ \downarrow & & \downarrow \cong & & \downarrow \cong \\ + & \xrightarrow{\text{Gaussian elimination}} & \mathcal{S}(\mathcal{M}, b_{\bullet}) & \xrightarrow{\text{normalization}} & \mathcal{N}(\mathcal{M}, b_{\bullet}, \{\mathbf{T}\}) \end{array}$$

We adopt Ruiz's reduction framework [2], which removes symmetries by reducing each realization to a normalized representative via:

- $\mathcal{S}(\mathcal{M}, b_{\bullet})$: Gaussian elimination on a fixed basis $b_{\bullet}$, giving standard form $(I \mid S)$.
- $\mathcal{N}(\mathcal{M}, b_{\bullet}, \{\mathbf{T}\})$: Normalize along a spanning forest $\{\mathbf{T}\}$ of the fundamental graph $\text{FG}(\mathcal{M}, b_{\bullet})$.

Example.

$$\begin{pmatrix} 1 & 2 & 3 & * & * & * \\ 3 & 2 & 1 & * & * & * \\ 2 & 1 & 3 & * & * & * \end{pmatrix} \xrightarrow{\text{Gaussian elimination}} \begin{pmatrix} 1 & 0 & 0 & 4 & 7 & 0 \\ 0 & 1 & 0 & 2 & * & 5 \\ 0 & 0 & 1 & 3 & * & * \end{pmatrix} \xrightarrow{\text{normalize}} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & * & 1 \\ 0 & 0 & 1 & 1 & * & * \end{pmatrix}$$

Proof Ingredients

- **NEO has non-rephasing real column:**

Standard realizations of NEOs always contain at least one column that cannot be globally rephased into reals.

Example.

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & \dot{2} & 1 \\ 0 & 0 & 1 & 1 & \dot{3} & \dot{i}i \notin \mathbb{R} \end{pmatrix}$$

- **Triangle Lemma (magnitude recovery):**

A single unknown magnitude $\dot{x}$ of $\text{ph}(\alpha \dot{x} + \beta y) = \gamma$ is determined if $\alpha, \beta, \gamma \in \mathbb{S}^1$ closes a non-degenerate triangle in phase.

Example.

$$\begin{pmatrix} 1 & 1 \\ 1 & \dot{i}\alpha \end{pmatrix} \rightsquigarrow \text{ph}(\dot{i} - 1) = \varphi(e_1, e_4, e_6) = \frac{\dot{i} - 1}{\sqrt{2}} \xrightarrow{\text{T.L.}} \dot{i} = 1$$

- **Uniform realization:**

$U_{d,d+2}$ realizations take the form

$$(I \mid 1 \mid \mathbf{q}),$$

with $\mathbf{q}$'s entries non-zero and distinct.

Entries marked with a dot (e.g., $\dot{i}, \dot{2}$) represent unknown scaling values.

Sharpness of the NEO+2U Criterion

The rank+2 uniformity condition is **sharp**: Realization spaces may fail to be singleton without this condition, even for rank+1 uniformity NEO phased matroids

Example. For all $r > 1$, M_r and M'_r each induce a fixed phased matroid $\mathcal{M} = \mathcal{M}(M_r)$ and $\mathcal{M}' = \mathcal{M}(M'_r)$, respectively, without singleton realization spaces:

$$M_r = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & r \\ 0 & 0 & 1 & 1 & \dot{i} & 0 \end{pmatrix} \quad M'_r = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & r \\ 0 & 0 & 1 & 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 3 & \dot{i} & 0 \end{pmatrix}$$